

Viatom File Parse Protocol

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1. Overview

this document shows how files are managed on the device and how to parse them.

2. File Parse

2.1 Blood Pressure Monitor (BP2/BP2A)

BP2/BP2A saves ECG measurements file and BP measurements file.

2.1.1 ECG files

every measurement saves by one ECG file. the file name format is "yyyyMMddhhmmss", like "20180601114631" means the measurement starts at "2018-06-01 11:46:31"

File Content

File Content	Description
File Header	File Head, include file info, diagnostic result
Waveform	125 Hz sample rate, every sample dot is 2 bytes

Waveform sample value transform to actual voltage value:

$$x(n) = 1.0035 * (n * 1800) / (4096 * 178.74)$$

```
File Header {
    unsigned char    file_version;           // e.g. 0x01 : v1
    unsigned char    file_type;              // 0: BP, 1: ECG 1: BP; 2: ECG
    unsigned int     measuring_timestamp;     // timestamp
    unsigned char    reserved1[4];           // reserved
    unsigned int     recording_time;
    unsigned char    reserved2[2];           // reserved
    unsigned int     result;                 // diagnostic result
    unsigned short   hr;                    // bpm
    unsigned short   qrs;                   // ms
    unsigned short   pvcs;
    unsigned short   qtc;                   // ms
    unsigned char    is_connect_cable;
```

```

    unsigned char    reserved3[19];           // reserved
}

```

ECG diagnostic result

value	situation	Description
0x00000000	regular	regular ECG
0xFFFFFFFF	poor waveform quality	unable analysis
0xFFFFFFF0	lead off	unable analysis
0x00000001	HR>100bpm	high HR
0x00000002	HR<50bpm	low HR
0x00000004	R-R irregular	irregular heart beat
0x00000008	PVC	PVC
0x00000010	Cardiac arrest	Cardiac arrest
0x00000020	Atrial Fibrillation	Atrial Fibrillation
0x00000040	QRS>120ms	QRS wide
0x00000080	QTc>450ms	QTc long
0x00000100	QTc<300ms	QTc short

2.1.2 BP files

every measurement saves by one BP file. the file name format is "yyyyMMddhhmmss", like "20180601114631" means the measurement starts at "2018-06-01 11:46:31"

File Content

```

Struct {
    unsigned char    file_version;           // e.g. 0x01 : v1
    unsigned char    file_type;             // 0: BP, 1: ECG 1: BP; 2: ECG
    unsigned int     measuring_timestamp;    // timestamp
    unsigned char    reserved1[4];          // reserved
    unsigned char    status_code;
    unsigned short   systolic_pressure;
    unsigned short   diastolic_pressure;
    unsigned short   mean_pressure;
    unsigned char    pulse_rate;
    unsigned char    medical_result;        // diagnostic result, bit0:irregular
    heart beat
    unsigned char    reserved2[19];         // reserved
}BP Record

```

2.2 ECG recorder (ER2)

ER2 saves ECG measurements files. every measurement has one file. the file name format is "RyyyyMMddhhmmss", like "R20180601114631" means the measurement starts at "2018-06-01 11:46:31"

File Content

File Content	Description
File Header	File Head, include file info
Waveform	125 Hz sample rate, every sample dot is 2 bytes
File Tail	File Tail

Waveform sample value transform to actual voltage value:

$$x(n) = 1.0035 * (n * 1800) / (4096 * 178.74)$$

```
File Header {
    Unsigned char file_version;    //e.g. 0x01 : v1
    Unsigned char reserved[9];    //reserved
}
```

```
File Tail {
    Unsigned int    recording_time;    //e.g. 3600 : 3600s
    Unsigned short  data_crc;          //CRC checksum
    Unsigned char   reserved[10];     //reserved
    Unsigned int    magic;             //fixed value: 0xA55A0438
}
```

2.3 VisualBeat (ER1)

ER2 saves ECG measurements files. every measurement has one file. the file name format is "RyyyyMMddhhmmss", like "R20180601114631" means the measurement starts at "2018-06-01 11:46:31"

File Content

File Content	Description
File Header	File Head, include file info
sample dot	saves every 1 second, every sample dot is 3 bytes
File Tail	File Tail

Waveform sample value transform to actual voltage value:

$$x(n) = 1.0035 * (n * 1800) / (4096 * 178.74)$$

```
File Header {
    Unsigned char file_version;    //e.g. 0x01 : v1
    Unsigned char reserved[9];    //reserved
}
```

```
sample dot {  
    Unsigned char hr;           //heart rate (30~250)  
    Unsigned char motion;       //Sum of 3-axis acceleration(0~255)  
    Unsigned char vibration;     //bit7: 1 => is vibration mark  
}
```

```
File Tail {  
    Unsigned int recording_time; //e.g. 3600 : 3600s  
    Unsigned short data_crc;     //CRC checksum  
    Unsigned char reserved[10];  //reserved  
    Unsigned int magic;          //fixed value: 0xA55A0438  
}
```